

Black Friday Sales & Marketing Analysis

Executive Summary Report

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Page 1 – Key Insights

Sales Performance

- Total Sales reached **Rs.5bn**, with an average purchase of **Rs.9.26K**.
- City B dominates sales, contributing **41.5%** of revenue.
- Male customers drive **76.7%** of purchases, showing gender imbalance.
- Age group **26–35** is the most profitable, contributing **Rs.2bn**.
- Product Category 7 leads with **Rs.1.91bn** sales, followed by Categories 5 & 6.
- Occupation 4 contributes disproportionately (**Rs.0.64bn**).

Hypothesis Testing Outcomes

- Gender vs Purchase: T-test confirmed **male customers spend significantly more** than female customers ($p < 0.05$).
- Age vs Purchase: ANOVA showed **age group differences are statistically significant**, with 26–35 outperforming all others.
- Occupation vs Purchase: Certain occupations (4, 5, 6) have **higher mean purchase values**, validated by ANOVA ($p < 0.05$).
- Marital Status vs Purchase: No statistically significant difference between single and married customers ($p > 0.05$).

Customer Behavior

- Repeat customers: 5.89K, with City C showing the highest loyalty (3.14K).
- Category breadth: Many customers purchase across **14–16 categories**, indicating strong cross-category engagement.
- Customer Lifetime Value (CLV): High-value segment averages **Rs.9.3K**, medium value **Rs.2.5K**.
- Top customer spent the most, while Product P00025442 is the most purchased item.

CLV Segmentation Table

Segment	Average CLV (Rs.)	Customer Count
High Value	9300	2,000
Medium Value	2500	3,500
Low Value	1200	391

Hypothesis Test Results Table

Test	p-value	Outcome
Gender vs Purchase (T-test)	0.000	Significant
Age vs Purchase (ANOVA)	0.001	Significant
Occupation vs Purchase (ANOVA)	0.000	Significant
Marital Status vs Purchase (T-test)	0.73	Not Significant

Page 2 – Strategic Recommendations

Targeted Campaigns

- City-specific marketing: Focus promotions in City B for growth, and City C for retention.
- Loyalty programs: Strengthen engagement in City C where repeat customers dominate.
- Cross-selling strategies: Promote bundles for customers with high category breadth (14–16 categories).

Product Strategy

- Prioritize Category 7 in marketing campaigns.
- Introduce personalized offers for top-spending occupations (Occupation 4, 5, 6).
- Expand gender-specific promotions to attract female buyers (currently only 23.3% share).

Hypothesis-Driven Actions

- Gender gap: Since hypothesis testing confirmed males spend significantly more, design **female-focused campaigns**.
- Age targeting: ANOVA validated 26–35 as the highest-spending group — allocate **premium offers and upsell strategies**.
- Occupation focus: Occupations 4, 5, 6 show statistically higher purchases — tailor **occupation-based loyalty rewards**.
- Marital status: With no significant difference, avoid segmenting campaigns by marital status — resources are better spent on age and occupation.

Business Impact

- Retention boost: Loyalty campaigns in City C could increase repeat purchases by 10–15%.
- Revenue growth: Cross-selling across categories can lift average CLV by 20%.
- Market balance: Gender-focused promotions may unlock untapped female customer potential.
- Validated targeting: Hypothesis testing ensures campaigns are **data-driven and statistically sound**, reducing wasted spend.

Suggested visuals: Hypothesis testing charts (t-test, ANOVA), Repeat Customers by City, CLV by Segment.